Floodplain (River / Rural / Dynamic Water & Terrain Conditions)

Executive Summary

Floodplain is a **river-rural hybrid battlefield** built for **dynamic water-level combat, shifting terrain conditions, amphibious mobility, and autonomous environmental adaptation.** The terrain features braided rivers, seasonal flood zones, rural farmlands, levees, wooden bridges, irrigation canals, wetlands, and small villages. The map emphasizes Obsidian Protocol's autonomous terrain-reactive logic, sensor-fusion under humidity and fog, comms-relay warfare, and logistics under constantly changing river conditions. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10×10 m physical area representing ~1 km² of rural river terrain. Key features include flood-driven chokepoints, amphibious routes, shifting mudflats, and dynamic cover created by noise or falling water.

Design Goals & Gameplay

Dynamic Water-Level Combat
Floodplain prioritizes combat where water levels rise and fall. Bridges become chokepoints, mudflats appear or vanish, and river channels shift mid-match.
Terrain-Reactive Movement
Heavy armor struggles in soft soil; light vehicles excel. Infantry and drones adapt to flooded fields, washed-out roads, and shifting wetlands.
Amphibious & Rural Operations
Players conduct river crossings, canal assaults, and farmland pushes. Rural structures provide cover but collapse under sustained fire.
Autonomy & Environmental Adaptation
Autonomous units choose routes based on water depth, soil stability, vegetation density, and sensor clarity. Units reroute when terrain becomes impassable.
Sensor & Information Warfare
Fog, humidity, river spray, and vegetation distort sensors. Relay towers on levees restore clarity. Destroying relays creates blind pockets across the floodplain.
Logistics & Resources
Fuel depots, farm silos, and river docks serve as resupply nodes. Convoys must navigate washed-out roads and avoid ambush-prone wetlands.
Deployment & Progress
A fixed deployment budget (~10,000 points) forces players to choose between amphibious units, rural infantry, or heavy armor. Blue deploys from the southwest river dock; Red deploys from the northeast levee stronghold.

Spatial Constraints (Nreal AR)

Device Capabilities
Nreal Light/Air ~52° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.
Playable Area & Scaling
10×10 m physical area ≈ 1 km² rural river zone (1:100 scale). Designed for tabletop AR.
Tracking & Occlusion
Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System
1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.
Key Locations
Blue HQ: River Dock Command1 — (0,0). 150×150 m Boats, amphibious vehicles, cargo crates, rural staging.
Red HQ: Levee Stronghold1 — (1000,1000). 150×150 m Elevated bunker with radar mast. AA emplacements, and flood-control systems.
Central Floodplain Basin — (500,450). 300×300 m Braided rivers, wetlands, mudflats, shifting terrain.
Farmstead Depot — (250,800). 120×120 m Barns, silos, fuel tanks, interior flanking routes.
Wetland Forest Network — (800,300). 200×200 m Dense vegetation, shallow water, stealth movement.
River Crossing Corridor — (0,450 → 400,500) Primary amphibious chokepoint: bridges, fords, and shallow water levels.
Rural Road Route — (300,0 → 450,350) Safer land route with partial cover.
Levee Relay A — (650,150). +260 m Long-range sensor overlook point.
Levee Relay B — (150,650). +260 m Comms relay hub and flood-monitoring node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

River Crossing Corridor
Major chokepoint; controlling it determines amphibious movement.
Floodplain Basin
Dynamic terrain; long-range engagements shift as water rises.
Wetland Forest Network
Interior flanking routes; stealthy but slow.
Rural Road Route
Predictable but safe; vulnerable to overwatch from levees.
Sightlines
Levee Relay A provides long-range sensor coverage (~300 m). Fog and humidity create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW river dock + amphibious staging.
Red Deployment: NE levee stronghold + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain
Rivers, wetlands, mudflats, levees, rural fields, farmsteads.
Vegetation
Grass, reeds, wetland trees, farmland crops.
Buildings & Props
Military: AA guns, radar dishes, bunkers. Rural: Barns, silos, houses, bridges. Logistics: Trucks, crates, fuel drums, boats, drones.
Effects
Fog, humidity haze, water spray, mud splashes.
Audio
River flow, insects, machinery, distant artillery, radio chatter.

LOD & Budget

Asset Cat	Count	Tri/Item	Total Tris
~~River/Mire~~	2	200k	600k
Rural/Ind	6	80k	320k
Buildings	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget
2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins
Unreal ARTemplate + Nreal SDK (NREAL). ARKit/ARCore enabled.
Rendering
Forward renderer, stationary lights, baked GI, aggressive culling.
Anchors
Single spatial anchor for entire map.
Interaction
Phone pointer for pan/zoom; HUD as screen-space widgets.
Multiplayer
Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones
1. Prototype Layout — Month 1 Block-out rivers, wetlands, farmsteads, AR anchor test.
2. Core Systems — Month 2 Dual-tech (land + water), amphibious autonomy, dynamic terrain logic.
3. AR Integration — Month 3 Anchors, plane detection, basic interactivity.
4. Content Fill — Month 4–5 Final models, textures, lighting, audio.
5. Gameplay & Polish — Month 6–7 Objectives, balance, VFX (fog, water spray, mud).
6. Testing & QA — Month 8 Multiplayer stress test, sensor validation.

Testing Plan

Autonomy
Verify amphibious routing, dynamic terrain adaptation, hazard avoidance.
Comms/Sensors
Test relay destruction, fog-of-war updates, LOS checks under humidity.
Performance
≥60 Hz AR render on target device.
QA Checklist
Navigation, dynamic terrain logic, deployment budget enforcement, anchor drift, sensor accuracy.